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# KNN IMPLEMENTATION
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# Step 1: Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
from sklearn.metrics import accuracy_score

# --------------------------------------------
# Step 2: Create Dataset
# --------------------------------------------

data = {
    'Age': [22,25,47,52,46,56,48,55,60,62],
    'Salary': [25000,27000,48000,52000,50000,
               62000,58000,65000,70000,72000],
    'Purchased': [0,0,1,1,1,1,1,1,1,1]
}

dataset = pd.DataFrame(data)

print(dataset)

# --------------------------------------------
# Step 3: Dataset Information
# --------------------------------------------

print("\nFirst 5 Rows")
print(dataset.head())

print("\nShape of Dataset")
print(dataset.shape)

print("\nDescription")
print(dataset.describe())

# --------------------------------------------
# Step 4: Split Features and Labels
# --------------------------------------------

X = dataset.iloc[:,0:2].values
y = dataset.iloc[:,2].values

# --------------------------------------------
# Step 5: Train Test Split
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# ------------------------------------------

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=0
)

# ------------------------------------------
# Step 6: Feature Scaling
# ------------------------------------------

sc = StandardScaler()

X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

# ------------------------------------------
# Step 7: Train KNN Model
# ------------------------------------------

classifier = KNeighborsClassifier(
    n_neighbors=3,
    metric='minkowski',
    p=2
)

classifier.fit(X_train, y_train)

# ------------------------------------------
# Step 8: Predictions
# ------------------------------------------

y_pred = classifier.predict(X_test)

print("\nPredicted Values")
print(y_pred)

# ------------------------------------------
# Step 9: Confusion Matrix
# ------------------------------------------

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix")
print(cm)

# ------------------------------------------
# Step 10: Classification Report
# ------------------------------------------

cr = classification_report(y_test, y_pred)

print("\nClassification Report")
print(cr)
```

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# ----------------------------------------
# Step 11: Accuracy
# ----------------------------------------

acc = accuracy_score(y_test, y_pred)

print("\nAccuracy :", acc)

# ----------------------------------------
# Step 12: Visualization
# ----------------------------------------

plt.scatter(dataset['Age'],
            dataset['Salary'],
            c=dataset['Purchased'])

plt.xlabel("Age")
plt.ylabel("Salary")
plt.title("KNN Dataset Visualization")

plt.show()
```